Data Science for Energy System Modelling
Lecture 7: Redispatch and Capacity Expansion

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Hybrid Format Information: Zoom and Recordings

The recordings will be **publicly available** in this ISIS course and may also be published on YouTube or other platforms.

- The camera and microphone are directed **only at the lecturer**.
- Student contributions are **not recorded**.
  When interaction takes place, the recording will be **paused**.
- If you **voluntarily speak** while the recording is running, you **consent** to the recording and publication of your contribution.
- You may **withdraw** this consent at any time for the future; that part will then be deleted or edited.
- Participants who wish to remain anonymous may join under a **pseudonym** and/or use the **chat or Q&A** function instead of speaking.

Recap: Short-Term Marginal Generation Cost (STMGC)

The short-term marginal generation cost of a generator s can be calculated from the sum of **fuel cost** (fc_s), the operation-dependent **operation & maintenance cost** (VOM_s), and potentially the cost **CO_2 emission certificates** γ ($\frac{€}{t_{CO2}}$).

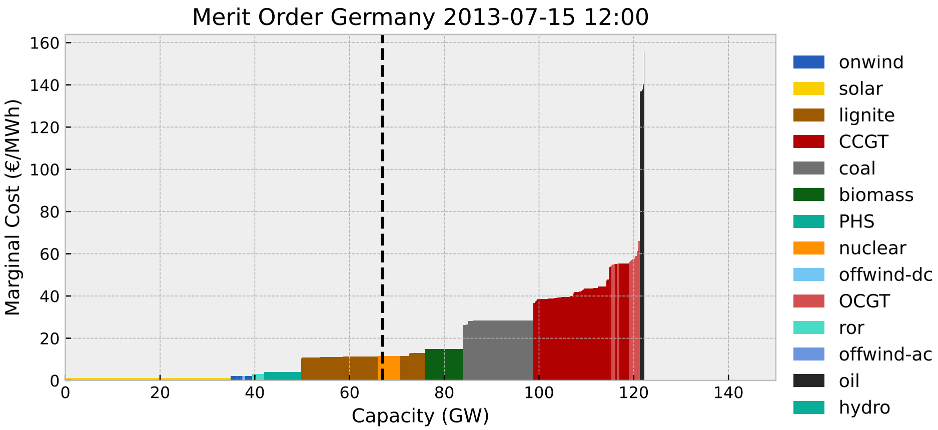
$$STMGC_s = o_s = \frac{fc_s}{\eta_s} + VOM_s + \gamma \frac{\rho_s}{\eta_s},$$

where ρ_s is are the specific emissions of the fuel used ($\frac{t_{CO2}}{MWh_{th}}$) and η_s is the conversion efficiency from thermal to electrical energy ($\frac{MWh_{el}}{MWh_{th}}$).

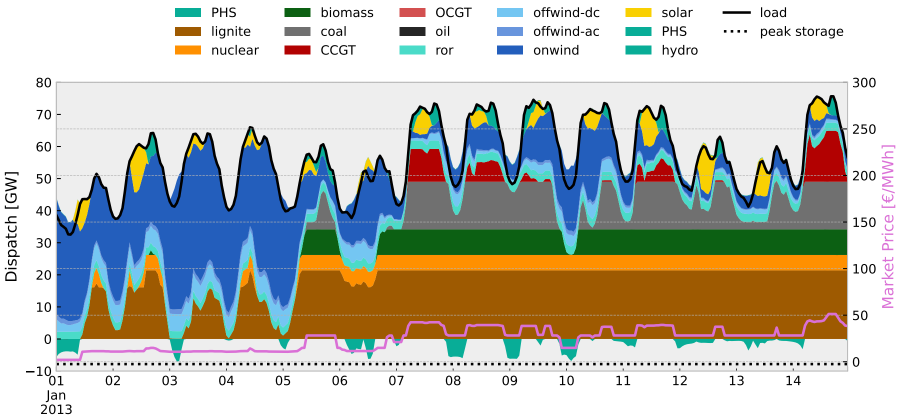
With aligned units:

$$\frac{€}{MWh_{el}} = \frac{\frac{€}{MWh_{th}}}{\frac{MWh_{el}}{MWh_{th}}} + \frac{€}{MWh_{el}} + \frac{€}{t_{CO2}} \frac{\frac{t_{CO2}}{MWh_{th}}}{\frac{MWh_{el}}{MWh_{th}}}$$

Recap: Merit order of generators

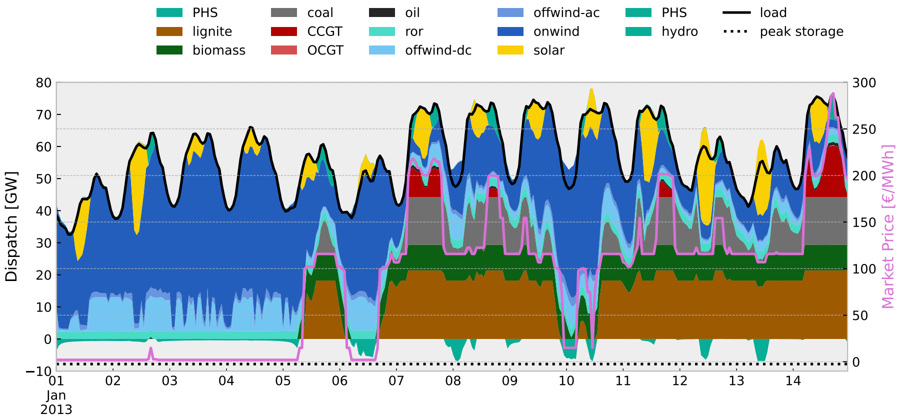


Cost-optimal dispatch & prices with current capacities over time



NB: no unit commitment constraints imposed on thermal generators (e.g. ramping, start-up cost, shut-down cost, minimum and maximum runtimes)!

Cost-optimal dispatch & prices with double wind + solar over time



NB: no unit commitment constraints imposed on thermal generators (e.g. ramping, start-up cost, shut-down cost, minimum and maximum runtimes)!

Analysis Example 1: Revenue, Expense, and Profit

We can calculate the generator revenues, expenses and profits.

Revenue = market price $\lambda_t \times$ electricity produced $g_{s,t}$:

$$\lambda_t g_{s,t}$$

Cost = STMGC $o_s \times$ electricity produced $g_{s,t}$:

$$o_s g_{s,t}$$

Profit = revenue – cost:

$$\lambda_t g_{s,t} - o_s g_{s,t} = (\lambda_t - o_s) g_{s,t}$$

Analysis Example 2: Carbon Intensity

The **carbon intensity** gives a measures of the CO_2 emissions per unit of electricity generated/consumed. The typical unit is **g/kWh**.

Carbon intensity of a generator s:

$$CI_s = \frac{\rho_s}{\eta_s}$$

Carbon intensity of a system at a given point in time t:

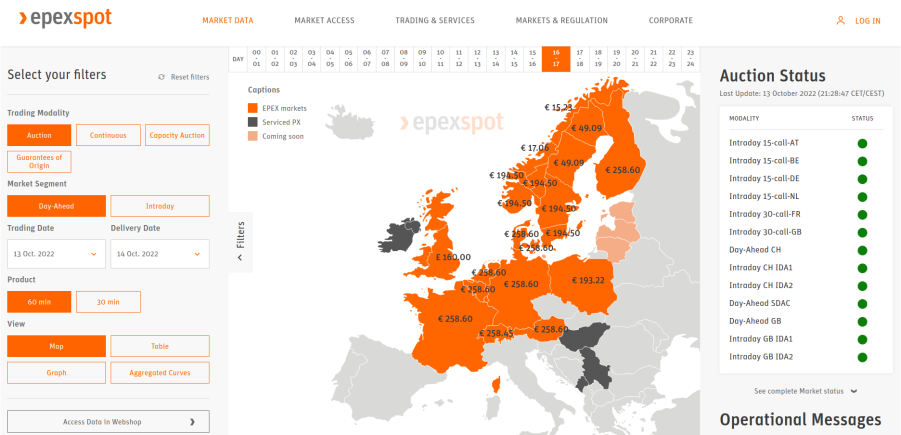
$$CI_t = \frac{\sum_s g_{s,t} \frac{\rho_s}{\eta_s}}{\sum_s g_{s,t}}$$

Carbon intensity for a system in total:

$$CI = \frac{\sum_{s,t} g_{s,t} \frac{\rho_s}{\eta_s}}{\sum_{s,t} g_{s,t}}$$

Checkout **live** carbon intensity: `https://app.electricitymaps.com/map`

Recap: Large bidding zones in Europe with market coupling



So far, we have only considered **isolated market zones** without cross-border trade.
We also did not consider **transmission constraints** within the bidding zones.

Source: `https://www.epexspot.com/en/market-data`

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Redispatch and curtailment for technical feasibility

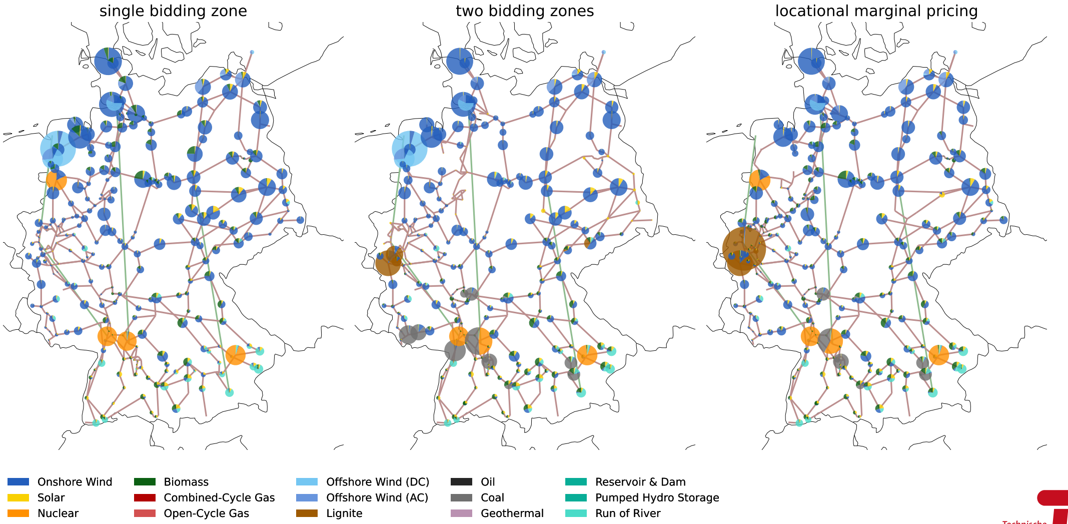
Market outcomes (dispatch and prices) are calculated **ignoring physical transmission constraints** within the bidding zones. However, the physical flows resulting from the market dispatch may **violate transmission limits**.

To safeguard the physical limits of the transmission network, the network operator may:

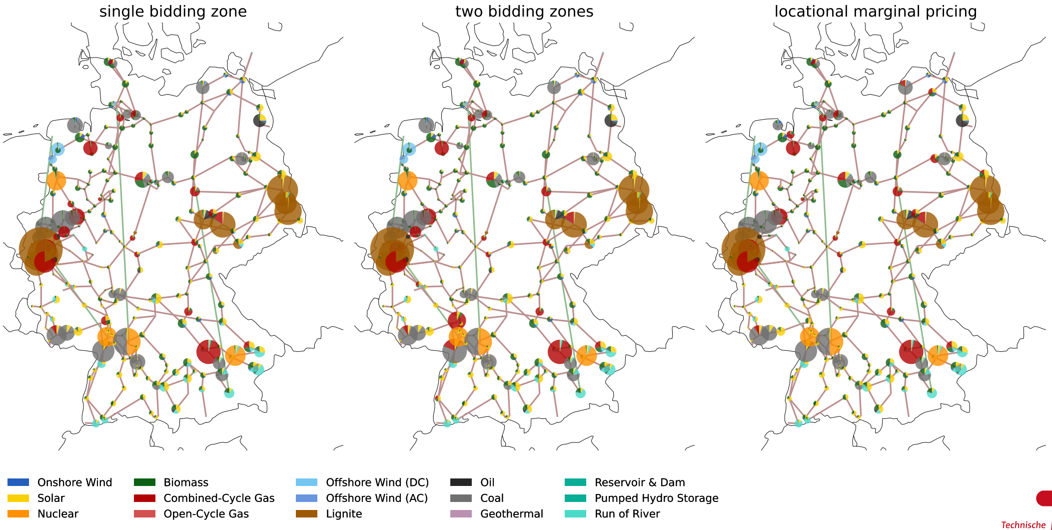
1. **curtail** renewables (primarily wind, but in principle also solar)
2. **re-dispatch** other power stations (ramp up, ramp down production of power plants)

This is generally **costly** and results in **lost carbon-free generation**.

Market outcomes under different bidding zones (Hour A, before redispatch)

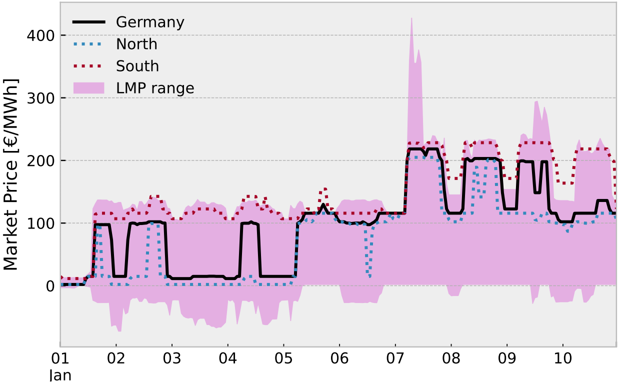


Market outcomes under different bidding zones (Hour B, before redispatch)



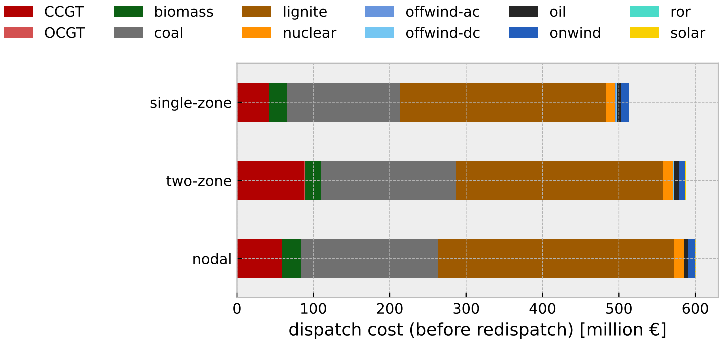
Market clearing prices under different bidding zones

Bidding zone splits **internalise** cross-zonal constraints in the market clearing process.



You can see quite nicely how market prices for Northern wind-rich and Southern demand-rich region **diverge** when market is split into two zones.

Operational costs before redispatch



Single-bidding zone appears to result in lowest cost, but it ignores **intrazonal** transmission constraints, i.e. it neglects transmission bottlenecks inside the bidding zone. Tackling these through **redispatch** measures will incur further costs.

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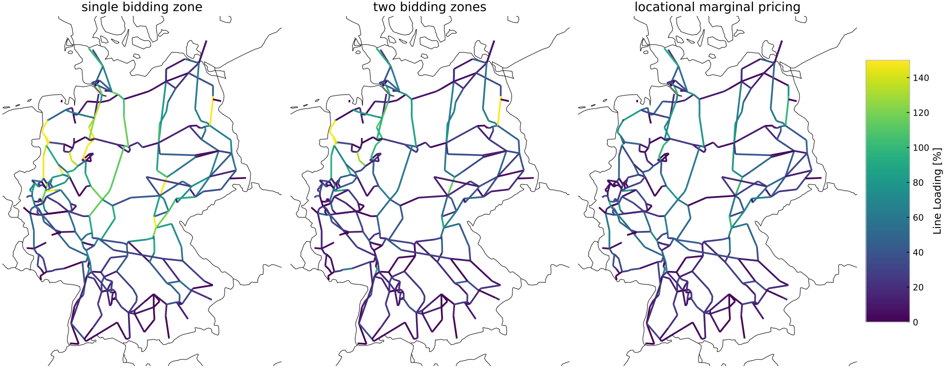
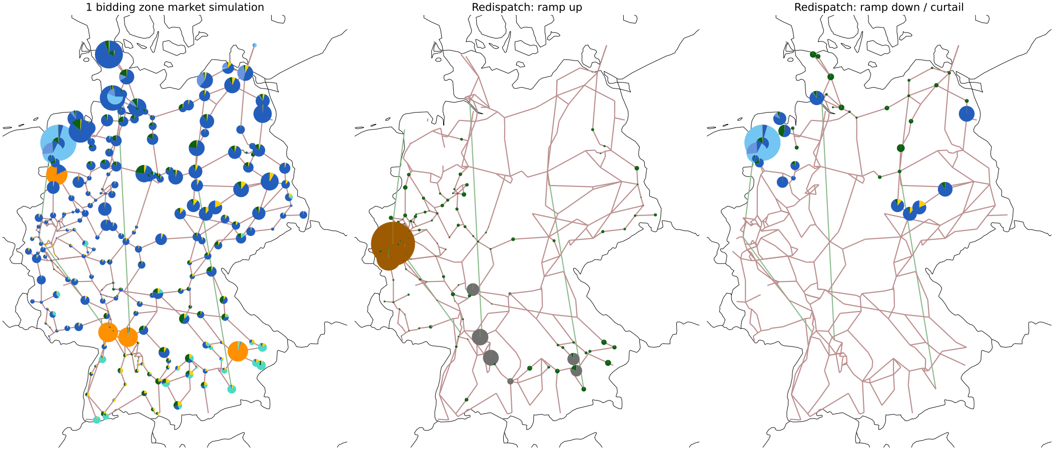
Line loading under different bidding zones (Hour A, before redispatch)



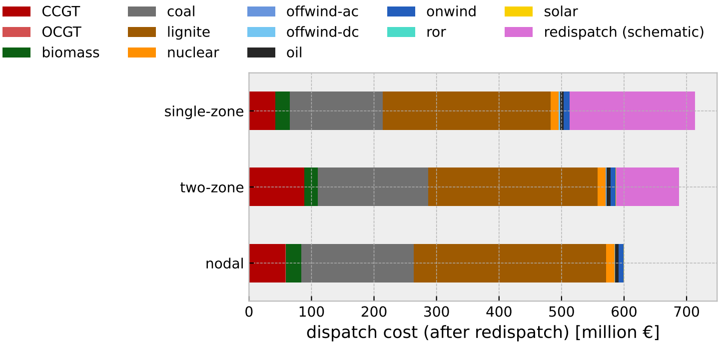
Single-bidding zone ignores **intrazonal** transmission constraints, i.e. its dispatch plan can lead to line overloadings that need to be avoided through **redispatch** measures. Locational marginal pricing has no line overloadings, as it manages grid bottlenecks already in the market and avoids them.

Redispatch in case of single bidding zone (Hour A)



Left: before redispatch **Centre:** redispatch ramp-up **Right:** redispatch ramp-down

Cost after redispatch (schematic)



Generally, locational marginal pricing will lead to the least costly outcome as it **directly internalises the basic physical constraints** of moving power around – dispatch is directly optimised around grid bottlenecks, not fixed retrospectively.

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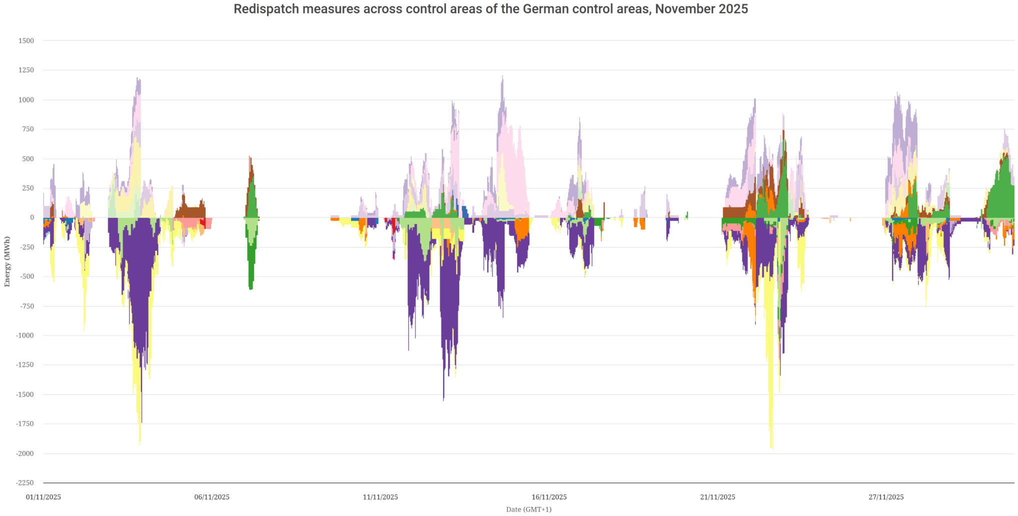
Redispatch volumes in November 2025 in Germany



Source: https://www.energy-charts.info/charts/power_redispatch/chart.htm

From modelling electricity markets to system planning

So far, we only looked at the **short-term perspective**, i.e. how to make the best use of the existing electricity system infrastructure.

Degrees of freedom / decision variables:

1. Dispatch of power plants
2. Operation of storage units
3. Curtailment of renewables

From modelling electricity markets to system planning

So far, we only looked at the **short-term perspective**, i.e. how to make the best use of the existing electricity system infrastructure.

Degrees of freedom / decision variables:

1. Dispatch of power plants
2. Operation of storage units
3. Curtailment of renewables
4. Investment in new generation
5. Investment in new storage units
6. Transmission reinforcement

Now, we will also look at the **long-term perspective**. We will optimise investment in the capacities of generators, storage and transmission by promoting the capacities, like G and F, to decision variables in the optimisation problem.

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Typical questions: What technology mix to build to balance VRE?

Build your own zero-emission electricity supply

Introduction

This tool calculates the cost of meeting a constant electricity demand from a combination of wind power, solar power and storage for different regions of the world.

First choose your location to determine the weather data for the wind and solar generation. Then choose your cost and technology assumptions to find the solution with least cost. Storage options are batteries and hydrogen from electrolysis of water.

Fun things to try out:

- remove technologies with the checkboxes, e.g. hydrogen gas storage or wind, and see system costs rise
- set solar or battery costs very low, to simulate breakthroughs in manufacturing

See also this Twitter thread for an overview of the model's features and capabilities.

This is a **toy model** with a **strongly simplified** setup. Please read the warnings below.

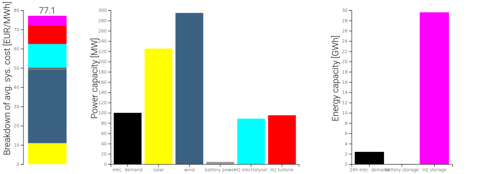
Step 1: Select location and weather year

- Select country
- Select state or province (for Australia, China, Germany, India, Russia and United States)
- Select point, rectangle or polygon, using the toolbox that appears at top-right

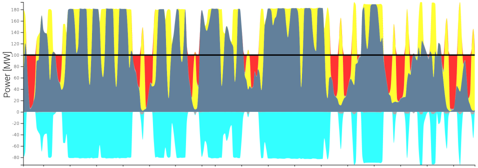
Average system cost [EUR/MWh]: 77.1

Cost is per unit of energy delivered in 2015 euros. For comparison, household electricity rates (including taxes and grid charges) averaged 211 EUR/MWh in the European Union in 2018 & 132 USD/MWh in the United States in 2019

Average marginal price of hydrogen [EUR/MWh LHV]: 78.8, [EUR/kg]: 2.60



Dispatch over a year to meet the constant demand (electricity demand is the black line; you can zoom and pan to see the details; negative values correspond to storage consuming electricity):



Source: https://model.energy/

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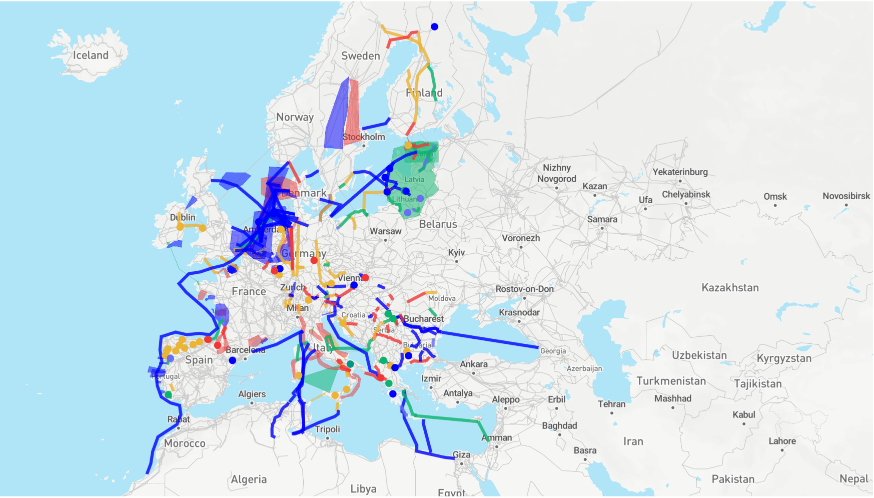
Typical questions: Where to reinforce transmission grid?



Source: `https://tyndp.entsoe.eu/`

Electricity system **operation** as optimisation problem (revision)

Decision variables (all operational) are shown in red. **Capacity expansion** is an extension of this problem!

$$\min_{g,e,f} \sum_{i,s,t} o_{i,s} g_{i,s,t} \quad \text{s.t.}$$

$$d_{i,t} = \sum_{s} g_{i,s,t} - \sum_{\ell} K_{i\ell} f_{\ell,t} \qquad \text{energy balance}$$

$$0 \leq g_{i,s,t} \leq \hat{g}_{i,s,t} G_{i,s} \qquad \text{generator limits}$$

$$0 \leq g_{i,s,t,\text{discharge}} \leq G_{i,s,\text{discharge}} \qquad \text{storage discharge limits}$$

$$0 \leq g_{i,s,t,\text{charge}} \leq G_{i,s,\text{charge}} \qquad \text{storage charge limits}$$

$$0 \leq e_{i,s,t} \leq E_{i,rs} \qquad \text{storage energy limits}$$

$$e_{i,r,t} = \eta^0_{i,s,t} e_{i,s,t-1} + \eta^1_s g_{i,s,t,\text{charge}} - \frac{1}{\eta^2_s} g_{i,s,t,\text{discharge}} \qquad \text{storage consistency}$$

$$-F_\ell \leq f_{\ell,t} \leq F_\ell \qquad \text{line limits}$$

$$0 = \sum_{\ell} C_{\ell c} x_\ell f_{\ell,t} \qquad \text{KVL}$$

Electricity system **planning** as optimisation problem

In addition to previous operational decision variables, we convert the parameters for capacities into investment decision variables with new associated parameters.

$$\min_{g,e,f,G,E,F} \sum_{i,s,t} w_t o_s g_{i,s,t} + \sum_{i,s} c_s G_{i,s} + c_{r,\text{dis/charge}} G_{i,r,\text{dis/charge}} + c_r E_{i,r} + c_\ell F_\ell \qquad \text{s.t.}$$

$$d_{i,t} = \sum_s g_{i,s,t} - \sum_\ell K_{i\ell} f_{\ell,t} \qquad \text{energy balance}$$

$$0 \leq g_{i,s,t} \leq \hat{g}_{i,s,t} G_{i,s} \qquad \text{generator limits}$$

$$0 \leq g_{i,r,t,\text{discharge}} \leq G_{i,r,\text{discharge}} \qquad \text{storage discharge limits}$$

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$$0 \leq e_{i,r,t} \leq E_r \qquad \text{storage energy limits}$$

$$e_{i,r,t} = \eta^0_{i,r,t} e_{i,r,t-1} + \eta^1_r g_{i,r,t,\text{charge}} - \frac{1}{\eta^2_r} g_{i,r,t,\text{discharge}} \qquad \text{storage consistency}$$

$$-F_\ell \leq f_{\ell,t} \leq F_\ell \qquad \text{line limits}$$

$$0 = \sum_\ell C_{\ell c} x_\ell f_{\ell,t} \qquad \text{KVL}$$

Changes in objective function with capacity expansion.

$$\min_{g,e,f,G,E,F} \sum_{i,s,t} w_t o_s g_{i,s,t} + \sum_{i,s} c_s G_{i,s} + c_{r,\text{dis/charge}} G_{i,r,\text{dis/charge}} + c_r E_{i,r} + c_\ell F_\ell$$

- We have **marginal cost** o_s for each generator (STMGC, €/MWh),
- we now also have **specific capital costs** $c_\star$ (e.g. building new power plant, €/MW),
- and a weighting w_t denoting how much time a time step t represents.
- Altogether, this objective function minimises **total system costs**.

Units of objective function with capacity expansion

$$\min \quad \overbrace{\sum_{i,s,t} \mathrm{h} \cdot \frac{€}{\mathrm{MWh}} \cdot \mathrm{MW}}^{\text{operational costs}} + \overbrace{\sum_{i,s} \frac{€}{MW} \cdot \mathrm{MW} + \frac{€}{MWh} \cdot \mathrm{MWh}}^{\text{investment costs}}$$

- For a fair comparison between operational costs and investments, we need to align the terms such that they **represent the same time frame**.
- **Example:** compare technology with high variable costs but low upfront cost versus technology with low variable cost and high upfront cost.
- **One common option: yearly system costs (€/a)**
  - operational: one year of representative operational conditions ($\sum_t w_t = 8760$h)
  - investment: annualised investment costs ($c_\star$ in €/MW/a = €/MW/8760h)

Present Value

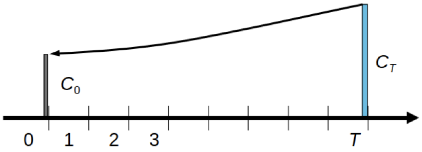
To allow comparison between income and costs in different years, we need to agree on a particular point in time to evaluate the cash flows; e.g. today's value, known as **present value**.

For an **interest rate** r, we multiply the cash flow in t years by the discount factor

$$\frac{1}{(1+r)^t}$$

to calculate the present value. We have **discounted** the future cash flow.

Future income or costs are **worth less** from today's point of view (if $r > 0$).



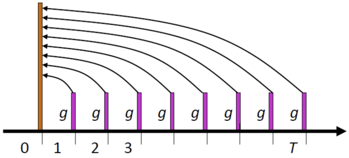
Source: Tom Brown

Present value factor for repeating cashflows

If we have a repeating constant cashflow CF for T years, the present value is calculated as follows:

$$PV = CF \cdot \sum_{t=1}^{T} \frac{1}{(1+r)^t}$$

The sum $\sum$ is called the **Present Value Factor** $PVF(r, T)$, which brings a series of equal cashflows (pink) to the present value (orange):



Source: For a derivation of the present value factor see `https://nworbmot.org/courses/ee-22/ee-6-financial_management.pdf`, slide 21.

Annuity factor to distribute upfront investments across lifetime

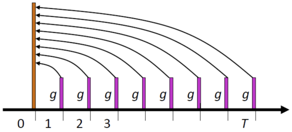
Vice versa, we can use the present value factor $PVF(r,T)$ to convert an initial investment I to identical annual payments to compare them with the cashflow in a year:

$$A = \frac{I}{PVF(r,T)}$$

A is also called the **annuity**, i.e. the annualised investment costs.

The **annuity factor** $a(r,T)$ spreads the capital cost I evenly over the operational years of the investment while taking account of the interest rate / discount rate r (like a mortgage for a house):

$$a(r,T) = \frac{1}{PVF(r,T)} = \frac{r}{1-(1+r)^{-T}}$$



reversed!

Source: Tom Brown

Annuity Factor: Table

Years	Interest rate [%] 3.0	3.5	4.0	4.5	5.0	6.0	7.0	8.0	9.0	10.0
1	1.030	1.035	1.040	1.045	1.050	1.060	1.070	1.080	1.090	1.100
2	0.523	0.526	0.530	0.534	0.538	0.545	0.553	0.561	0.568	0.576
3	0.354	0.357	0.360	0.364	0.367	0.374	0.381	0.388	0.395	0.402
4	0.269	0.272	0.275	0.279	0.282	0.289	0.295	0.302	0.309	0.315
5	0.218	0.221	0.225	0.228	0.231	0.237	0.244	0.250	0.257	0.264
6	0.185	0.188	0.191	0.194	0.197	0.203	0.210	0.216	0.223	0.230
7	0.161	0.164	0.167	0.170	0.173	0.179	0.186	0.192	0.199	0.205
8	0.142	0.145	0.149	0.152	0.155	0.161	0.167	0.174	0.181	0.187
9	0.128	0.131	0.134	0.138	0.141	0.147	0.153	0.160	0.167	0.174
10	0.117	0.120	0.123	0.126	0.130	0.136	0.142	0.149	0.156	0.163
11	0.108	0.111	0.114	0.117	0.120	0.127	0.133	0.140	0.147	0.154
12	0.100	0.103	0.107	0.110	0.113	0.119	0.126	0.133	0.140	0.147
13	0.094	0.097	0.100	0.103	0.106	0.113	0.120	0.127	0.134	0.141
14	0.089	0.092	0.095	0.098	0.101	0.108	0.114	0.121	0.128	0.136
15	0.084	0.087	0.090	0.093	0.096	0.103	0.110	0.117	0.124	0.131
20	0.067	0.070	0.074	0.077	0.080	0.087	0.094	0.102	0.110	0.117
25	0.057	0.061	0.064	0.067	0.071	0.078	0.086	0.094	0.102	0.110
30	0.051	0.054	0.058	0.061	0.065	0.073	0.081	0.089	0.097	0.106
35	0.047	0.050	0.054	0.057	0.061	0.069	0.077	0.086	0.095	0.104
40	0.043	0.047	0.051	0.054	0.058	0.066	0.075	0.084	0.093	0.102
45	0.041	0.044	0.048	0.052	0.056	0.065	0.073	0.083	0.092	0.101
50	0.039	0.043	0.047	0.051	0.055	0.063	0.072	0.082	0.091	0.101

Summary: optimising for lowest total annual system costs

$$\min_{g,e,f,G,E,F} \sum_{i,s,t} w_t o_s g_{i,s,t} + \sum_{i,s} c_s G_{i,s} + c_{r,\text{dis/charge}} G_{i,r,\text{dis/charge}} + c_r E_{i,r} + c_\ell F_\ell$$

$$\min \overbrace{\sum_{i,s,t} \mathrm{h} \cdot \frac{€}{\mathrm{MWh}} \cdot \mathrm{MW}}^{\text{operational costs}} + \overbrace{\sum_{i,s} \frac{€}{MW} \cdot \mathrm{MW} + \frac{€}{MWh} \cdot \mathrm{MWh}}^{\text{annual investment costs}}$$

- **One common option: yearly system costs (€/a)**
  - operational: one year of representative operational conditions ($\sum_t w_t = 8760\text{h}$)
  - investment: annualised investment costs ($c_\star$ in €/MW/a = €/MW/8760h)

Additional constraints: Limits on CO_2 emissions

To investigate the transition to highly renewable systems, me might want to add a constraint on the total CO_2 emissions:

$$\sum_{i,s,t} \frac{\rho_s}{\eta_s} g_{i,s,t} \leq \Gamma \quad \leftrightarrow \quad \mu_{CO2}$$

where

- ρ_s are the specific CO_2 emissions of technology s per unit of fuel (tCO2/MWh) and
- η_s is the efficiency of the generator (i.e. ratio between thermal and electrical energy).
- Γ could correspond to e.g. political targets for CO_2 reduction (tCO2)

What does μ_{CO2} represent? What is the unit?

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ELSEVIER

The benefits of cooperation in a highly renewable European electricity network

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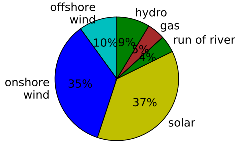
ABSTRACT

To reach ambitious European CO_2 emission reduction targets, most scenarios of future European electricity systems rely on large shares of wind and solar photovoltaic power generation. We interpolate between two concepts for balancing the variability of these renewable sources: balancing at continental scales using the transmission grid and balancing locally with storage. This interpolation is done by systematically restricting transmission capacities from the optimum level to zero. We run techno-economic cost optimizations for the capacity investment and dispatch of wind, solar, hydroelectricity, natural gas power generation and transmission, as well as storage options such as pumped-hydro, battery, and hydrogen storage. The simulations assume a 95% CO_2 emission reduction compared to 1990, and are run over a full historical year of weather and electricity demand for 30 European countries. In the cost-optimal system with high levels of transmission expansion, energy generation is dominated by wind (65%) and hydro (15%), with average system costs comparable to today's system. Restricting transmission shifts the balance in favour of solar and storage, driving up costs by a third. As the restriction is relaxed, 85% of the cost benefits of the optimal grid expansion can be captured already with only 44% of the transmission volume.

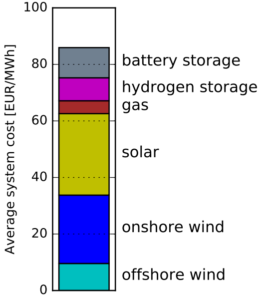
Source: https://doi.org/10.1016/j.energy.2017.06.004

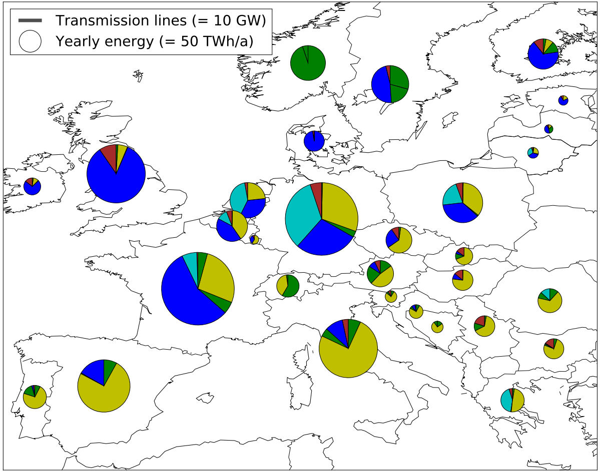
Costs: No interconnecting transmission allowed

Technology by energy:



Average cost **€86/MWh**:

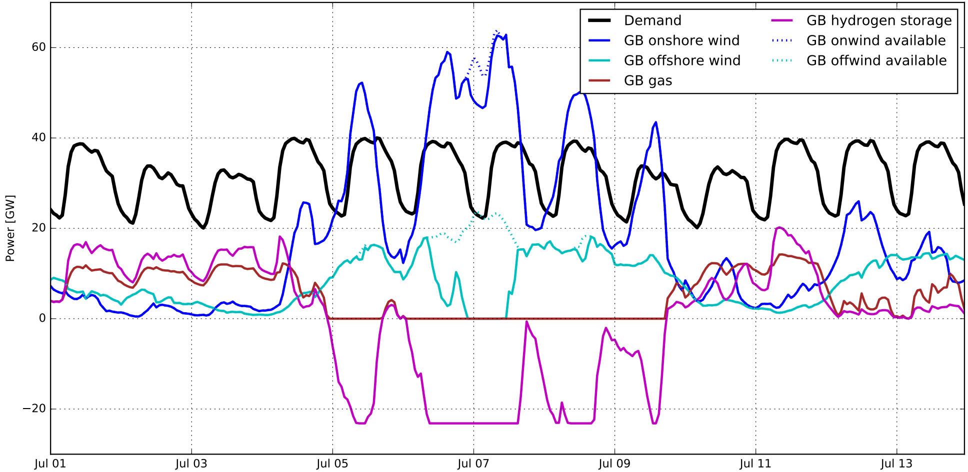




Countries must be self-sufficient at all times; lots of storage and some gas to deal with fluctuations of wind and solar.

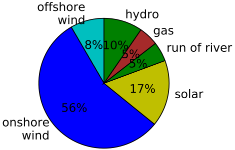
Dispatch with no interconnecting transmission

For Great Britain with no interconnecting transmission, excess wind is either stored as hydrogen or curtailed:

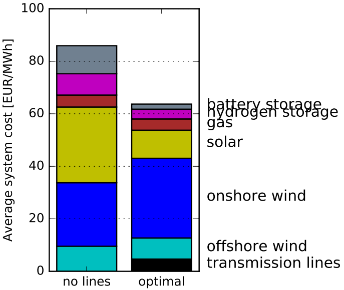


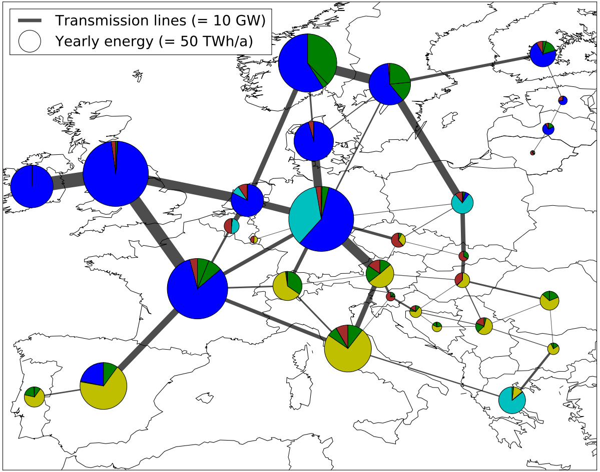
Costs: Cost-optimal expansion of interconnecting transmission

Technology by energy:



Average cost **€64/MWh**:





Large transmission expansion; onshore wind dominates. This optimal solution may run into public acceptance problems.

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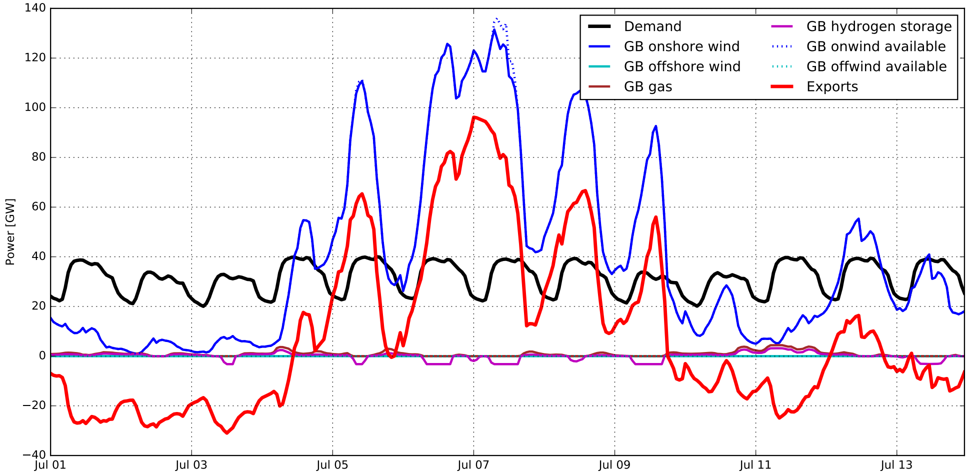
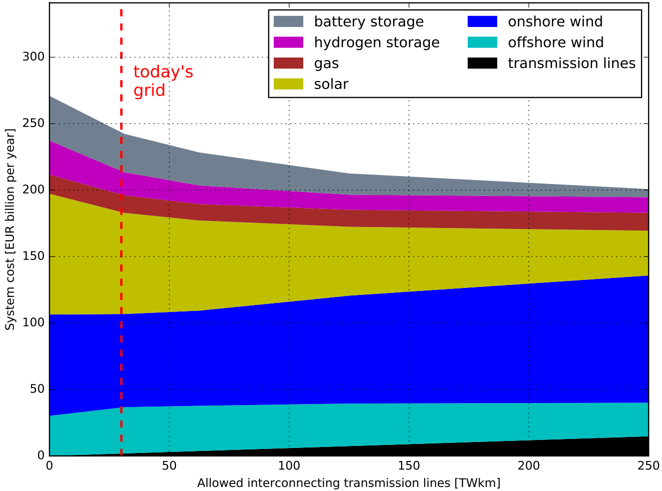
Almost all excess wind can be now be exported:



Electricity Only Costs Comparison



- Average total system costs can be as low as € 64/MWh
- Energy is dominated by wind (64% for the cost-optimal system), followed by hydro (15%) and solar (17%)
- Restricting transmission results in more storage to deal with variability, driving up the costs by up to 34%
- Many benefits already locked in at a few multiples of today's grid

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The near-optimal feasible space of a renewable power system model

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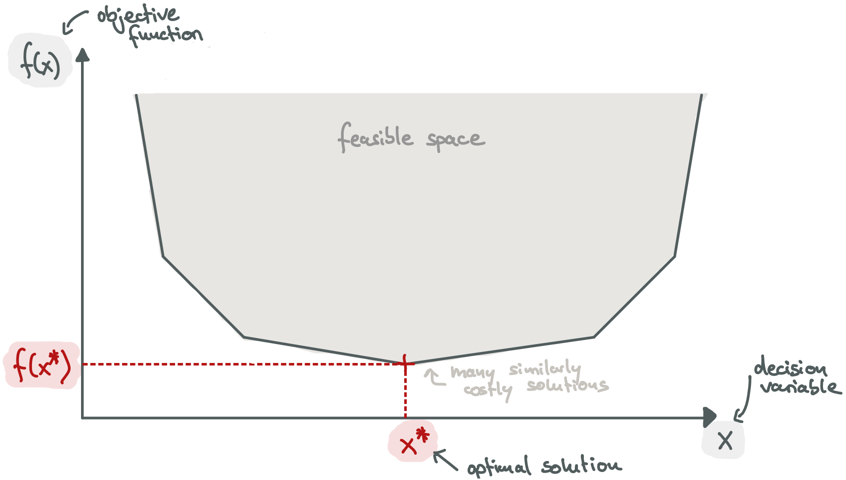
ABSTRACT

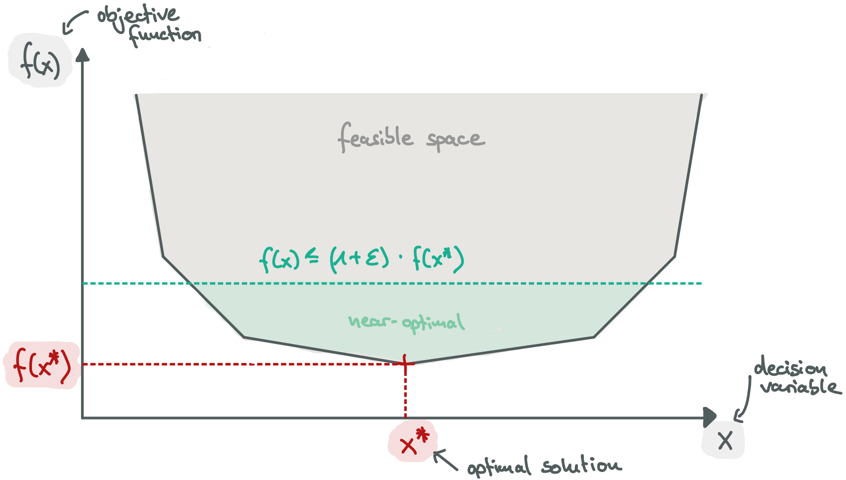
Models for long-term investment planning of the power system typically return a single optimal solution per set of cost assumptions. However, typically there are many near-optimal alternatives that stand out due to other attractive properties like social acceptance. Understanding features that persist across many cost-efficient alternatives enhances policy advice and acknowledges structural model uncertainties. We apply the modeling-to-generate-alternatives (MGA) methodology to systematically explore the near-optimal feasible space of a completely renewable European electricity system model. While accounting for complex spatio-temporal patterns, we allow simultaneous capacity expansion of generation, storage and transmission infrastructure subject to linearized multi-period optimal power flow. Many similarly costly, but technologically diverse solutions exist. Already a cost deviation of 0.5% offers a large range of possible investments. However, either offshore or onshore wind energy along with some hydrogen storage and transmission network reinforcement appear essential to keep costs within 10% of the optimum.

Source: https://doi.org/10.1016/j.epsr.2020.106690

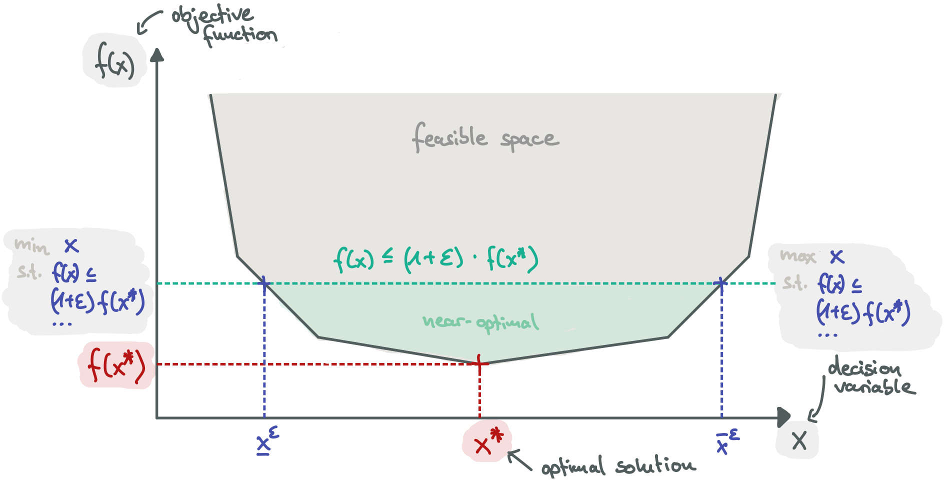
Large ‘degeneracy’ of possible energy systems near optimum.



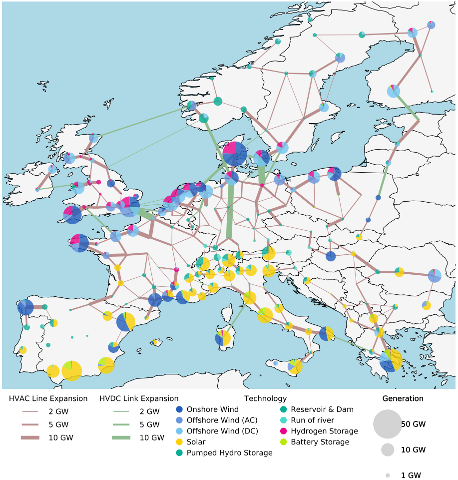
Consider the part of the feasible space within ε of the optimum $f(x^\star)$.



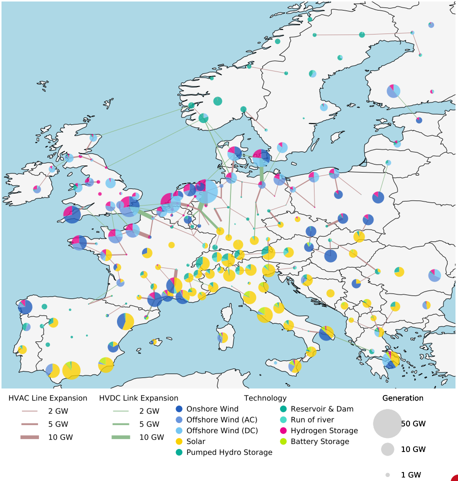
Now, within ε of the optimum $f(x^\star)$, minimise or maximise x.



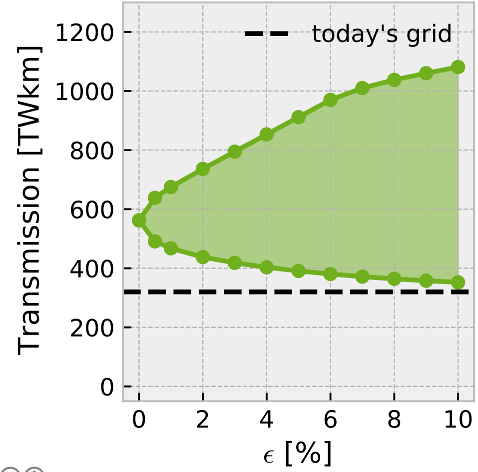
Least-Cost Transmission Volume $\varepsilon = 0.0\%$



Minimise Transmission Volume $\varepsilon = 5.0\%$

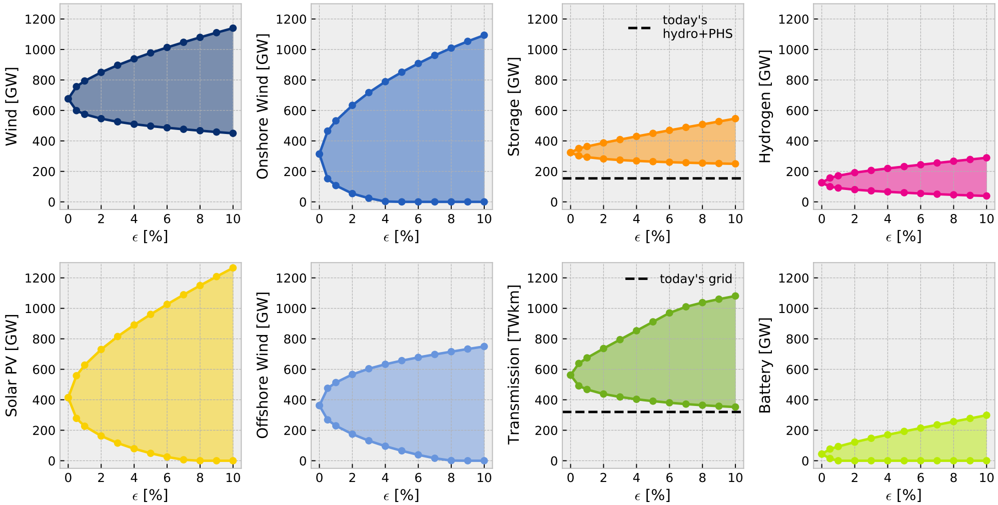


Pareto fronts exhibit a broad range of nearly cost-optimal solutions for transmission reinforcement.



- ϵ states %-deviation from least-cost system
- We minimise (lower curve) or maximise (upper curve) the transmission reinforcement while increasing the allowed cost penalty ϵ
- One way of **multi-objective optimisation** (minimise system costs, minimise transmission): ϵ-constraint method
- Curves are **Pareto fronts** denoting the set of solutions that cannot be improved in one objective without degrading the other

Flatness allows choosing solutions with higher public acceptance.



Up next

Next time we meet on **Thursday 08.01.2026** – no lectures on 18.12.2025!

1. Workshop on capacity expansion in `pypsa`.
2. Consultation hour for **Assignment 3**.